

Lab-Scale Lipid Nanoparticle (LNP) Manufacturing

Project Document

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Purpose of this document: This document records what has been done on this project to date, why each decision was made, what data/resources exist, and what the open problems and recommended next steps are.

1. Project Overview

The goal of this project is to build a lab-scale, low-cost manufacturing setup for lipid nanoparticles (LNPs) using microfluidic mixing. LNPs are the delivery vehicles used for mRNA vaccines and other nucleic-acid therapeutics, and their manufacture at lab scale conventionally relies on expensive commercial microfluidic systems (e.g. NanoAssemblr). The aim of this project is to replicate the core functionality of such a system, controlled multi-stream mixing of an aqueous phase and a lipid/ethanol phase using open-source hardware and 3D-printed parts, so that LNP formulation work can be done in-house at a fraction of the cost.

The overall system has two halves that had to be built in parallel:

- A set of syringe pumps that can deliver precise, controllable, pulsation-free flow rates for each inlet stream.
- A mixer geometry (the actual mixing chamber) that brings the streams together under turbulent/vortex conditions so that nanoprecipitation happens reproducibly.

Below is a chronological account of the work, organized by workstream.

2. Process Simulation Model and Process Flow Diagram

Early in the project, a process simulation model was built to map out the overall LNP manufacturing process conceptually (feed streams, mixing step, downstream dialysis/purification, and characterization steps) before committing to hardware. The process simulation model is there in the `final_zip.zip` file and you can find it in the google drive. There is a detailed user manual in it which will give complete detail about the model.

A process flow diagram (PFD) of the full process was also created, drawn using draw.io. This PFD is the single-page reference for how material flows through the system end-to-end (solvent/antisolvent prep → pumps → mixer → collection → downstream processing) and should be consulted first by anyone new to the project to understand where the hardware described in this document sits in the overall workflow. It is also attached in the google drive.

3. Process Analytical Technology (PAT) Research

Before finalizing the hardware approach, the team researched Process Analytical Technology (PAT), the general framework (promoted heavily by FDA/ICH guidance for pharmaceutical manufacturing) of building measurement and control directly into a process rather than relying purely on end-product testing. For LNP manufacturing specifically, this means thinking about in-line or at-line monitoring of things like flow rate, particle size (e.g. via in-line DLS), and mixing quality, rather than only characterizing the final product after the fact.

This research shaped two decisions downstream: (1) the emphasis on rigorously calibrating flow rate at the pump level rather than trusting motor step counts blindly, and (2) keeping the mixer design modular enough that in-line sensors could plausibly be added later (e.g. at the outlet tubing).

4. Syringe Pump Development

4.1 Reference design: Poseidon Github Repository

The syringe pump hardware is based on the open-source Poseidon syringe pump project. This was chosen over designing a pump from scratch because it is a mature, well-documented open-source design with an existing STL/CAD library and firmware base, which significantly reduced development time.

Repository link: <https://github.com/pachterlab/poseidon/tree/release>

4.2 Fabrication

- The Poseidon pump body and carriage parts were 3D printed in the Tinkerer's Lab.
- Remaining mechanical/electrical components (leadscrews, couplers, NEMA stepper motors, linear rods/bearings, CNC Shield V3 boards, A4988 stepper drivers, Arduino Uno/Nano boards, CH340-based USB-serial boards) were ordered. We have attached the purchase list in the google drive for further reference.
- Four pump units were assembled in total, following the Poseidon assembly instructions, to eventually supply the four inlets of the MIVM. There is a detailed video in the Poseidon repository for the assembly.
- NOTE: In google drive we have attached the modified base for syringe pump model as earlier it was facing screw fixing issues however for other parts to print you have to follow the Poseidon Github Repository.

4.3 Control electronics and firmware

Each pump is driven by an Arduino + CNC Shield V3 + A4988 driver stack, controlled using the AccelStepper library for smooth, non-blocking acceleration/deceleration motion profiles. Firmware was written so that a controlling PC can send serial commands (move-to targets, speed) and the pump executes them. The code for Arduino is attached in the google drive (for only X and Z axis of CNC shield and for all four connection you need to repeat the same code for Y and A axis).

Two of the four pumps have been calibrated so far (see Section 5); the other two are built but calibration is still pending because those two motors were not good.

5. Syringe Pump Calibration

Calibration methodology: flow rate was determined gravimetrically rather than by trusting the theoretical steps-per-mL conversion computed from syringe diameter and leadscrew pitch. This mirrors the calibration approach described in the referenced eNeuro paper (ENEURO.0240-19.2019), which was used as a methodological reference for how to rigorously validate syringe-pump flow rate against motor step rate.

The general procedure used for each calibration run:

- The acceleration and the maximum speed parameters (steps/sec² and steps/sec, set in the AccelStepper firmware) were kept fixed for a given pump; the actual run speed for each trial was set at or below this ceiling.
- The pump was run to dispense water into a pre-weighed vessel.
- The mass of water dispensed was measured directly using a weighing machine.
- The time taken to dispense that water was measured independently using a stopwatch.
- Flow rate was then calculated from these two measurements using the mass–density–volume relation (volume = mass / density of water, at ~0.976 g/mL, temperature-adjusted) and flow rate = volume / time.
- This was repeated across a range of run speeds, from ~420 steps/sec up to ~1050 steps/sec, to build a speed → flow-rate calibration curve for each pump.

The calibration data for both the pumps is attached in the google drive in the form of excel sheets.

Observed working range: at the acceleration setting used (200 steps/sec²), motor speeds from ~420 to ~1050 steps/sec produced flow rates from roughly 63 mL/min to 140 mL/min on the calibrated pumps. The relationship between speed and flow rate is close to linear over this range but should be fitted properly (linear regression) rather than assumed, and re-validated whenever syringe diameter, leadscrew, or tubing is changed.

Note: the literature-recommended per-inlet flow rate range for MIVM operation, per Markwalter & Prud'homme (2018) and Zhang & Clapp (2014), is 20–125 mL/min; the target total flow rate (TFR) for our MIVM is 250–300 mL/min across four inlets. This means the pumps need to be run well down their speed range (not near the top of the ~420–1050 steps/sec range tested) to hit realistic per-inlet flow rates for LNP formulation trials.

Important limitation: calibration in the literature-relevant 20–125 mL/min range has not yet been completed. At the lower speeds this range corresponds to, the pump produced noticeable jerking. The root cause identified was that the

syringe available was slightly larger than the pump model's syringe holder was designed for, so it had to be forced into place; this small dimensional mismatch is what produces the jerking, and it becomes more noticeable at lower speeds. The fix identified (but not yet carried out) is to rub the syringe holder area of the printed pump model by sanding or filing it down with sandpaper or a metal file so the minor dimensional excess is removed and the syringe seats properly without being forced. Once this is done, the pump should run smoothly at lower speeds, and calibration in the actual 20–125 mL/min operating range should be carried out as a priority.

A separate vibration issue was also observed with the pump body during operation. This was resolved (as a practical fix, not a design fix) by applying double-sided tape to the underside of the pump base and mounting the pump on a rubber sheet, which damps the vibration. This fix should be replicated across all four pump units.

6. Micromixer Development

6.1 Initial approach: Y/T-mixer

The first mixer concept explored was an open-groove Y-mixer / T-mixer design. This was abandoned because it requires microchannels (sub-mm scale) to achieve adequate mixing, and the campus 3D printers available to the team could not reliably resolve channels at that scale, prints either failed to resolve the channel geometry or produced channels too rough/blocked to be usable.

6.2 Shift to MIVM (Multi-Inlet Vortex Mixer)

Given the fabrication constraint above, the design was shifted to a Confined Impinging Jets (CIJ) / Multi-Inlet Vortex Mixer (MIVM) design, which achieves rapid turbulent mixing in a larger, more printable chamber (mm-scale rather than sub-mm channels), while still giving the mixing intensity needed for nanoprecipitation.

The design mainly follows: Markwalter, C.E. & Prud'homme, R.K., "Design of a Small-Scale Multi-Inlet Vortex Mixer for Scalable Nanoparticle Production and Application to the Encapsulation of Biologics by Inverse Flash NanoPrecipitation," J. Pharm. Sci. (2018), the supplementary information document supplied with this handover (1-s2_0-S0022354918303071-mm1.docx).

6.3 Scaling the dimensions

The supplementary information provides dimensions for the small μ MIVM design (chamber diameter 0.5 cm, channel height/width 0.075 cm each), plus dimensions for the original, larger benchtop MIVM (chamber diameter 0.6 cm, channel height 0.15 cm). It also gives the scaling relation used to relate chamber diameter to channel width across mixer sizes:

$$Dc_1^2 / w_1 = Dc_2^2 / w_2 \quad (\text{Eqn. S1, Markwalter \& Prud'homme 2018})$$

where subscript 1 refers to the original MIVM and subscript 2 refers to the scaled (μ MIVM-style) design. This relation was used to back-calculate a self-consistent set of dimensions for our printable mixer, since the paper's exact μ MIVM channel width (0.75 mm) was still below our printer's reliable minimum feature size.

The Autodesk file for the mixer model is attached in the google drive and you can we have also made some variations in a few dimensions which you can see from the 3D model.

Target operating point for this geometry: total flow rate (TFR) of 250–300 mL/min split across the four inlets.

6.4 Fabrication and materials

The mixer was 3D printed in PETG. PETG was chosen after researching solvent compatibility with pure ethanol (the typical organic/lipid-carrying solvent stream), drawing on Kaptan (2025), Grzelak (2021), Roberson (2019), Plaskolite's TEC409 chemical resistance data, and ORNL/TM-2012/88: PETG shows good short-to-medium-term resistance to ethanol at the concentrations and contact times relevant here.

7. Current Status and Open Issues

7.1 Syringe pump hardware/firmware

- Only 2 of 4 pumps are calibrated, and even that calibration only covers ~420–1050 steps/sec — it does not yet cover the literature-relevant 20–125 mL/min per-inlet range, because of the syringe-fit jerking issue described in Section 5. Fixing the syringe holder fit (sanding/filing) and then calibrating at lower speeds is a priority open task.
- The acceleration and the deceleration part is not accounted for in the calibration. That means the pump does not operate at constant speed throughout. Instead, it first accelerated, then moved at constant speed, and then it decelerated to stop. So, this acceleration/deceleration time needs to be calibrated.

7.2 Mixer

- There are leakages observed in the mixer model which need to be solved by applying a proper sealant.

8. Key References and Resources

- [Poseidon open-source syringe pump project \(GitHub\)](#): mechanical/firmware base for all four pumps.
- [eNeuro paper, ENEURO.0240-19.2019](#): reference for gravimetric syringe pump calibration methodology.
- Markwalter, C.E. & Prud'homme, R.K. (2018), "Design of a Small-Scale Multi-Inlet Vortex Mixer for Scalable Nanoparticle Production and Application to the Encapsulation of Biologics by Inverse Flash NanoPrecipitation," Journal of Pharmaceutical Sciences: primary reference for MIVM geometry and scaling relation (Eqn. S1). Supplementary info attached as 1-s2_0-S0022354918303071-mmc1.docx.
- Zhang, C. & Clapp, T. (2014): cross-reference for MIVM per-inlet flow rate operating ranges.
- Kaptan (2025), Grzelak (2021), Roberson (2019), Plaskolite TEC409, ORNL/TM-2012/88: PETG chemical/solvent-compatibility references used to justify PETG as the mixer print material.
- Process flow diagram: draw.io
- Process simulation model
- Raw calibration data: calibration_model_C.xlsx and calibration_model_unlabelled.xlsx